

Tutorial

Modern Physics

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For exercises from the textbook, the corresponding number is reported. Text may have been edited, for clarity.
Unless otherwise specified, all results are given with three significant digits, avoiding rounding errors.

The solutions are provided on an "as is" basis, without warranty. Any error report is appreciated.

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WEEK I

CHAPTER 3. THE EXPERIMENTAL BASIS OF QUANTUM PHYSICS

• Thompson experiment / Example 3.1

In an experiment similar to Thomson's, we use deflecting plates $\ell = 5.0$ cm in length with an electric field $E = 1.2 \times 10^4$ V/m. Without magnetic field, we find an angular deflection of $\theta = 30^\circ$, and with a magnetic field $B = 8.8 \times 10^{-4}$ T we find no deflection. What is the initial velocity of the electron and its charge to mass ratio q/m ?

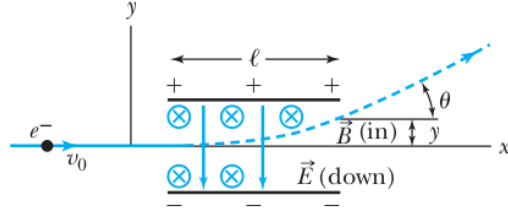


Figure 1: Thomson's experiment: the ratio e/m can be determined from E , B , θ , and ℓ .

Solution

The values of E and B for which there is no deflection determine the electron's velocity v_0

$$\vec{F} = q\vec{E} + q\vec{v} \times \vec{B} = 0$$

so that

$$v_x = \frac{|E|}{|B|} = v_0 = 1.4 \times 10^7 \text{ m/s.}$$

The deflection angle is determined by the following (geometric) relation

$$\tan(\theta) = \frac{v_y}{v_x},$$

and corresponds to the case with zero magnetic field, therefore, the velocity v_y is given by

$$v_y = a_y t = \frac{qE}{m} \frac{\ell}{v_0}.$$

where $t = \ell/v_0$ is the time for the electron to traverse the capacitor. Combining the above relations yields q/m as a function of E , B , θ , and ℓ

$$\frac{q}{m} = \frac{E \tan(\theta)}{B^2 \ell} = 1.8 \times 10^{11} \text{ C/kg.}$$

• **Energy of a light pulse / Ex. 3.38**

The human eye is sensitive to a pulse of light containing as few as $N_\gamma = 100$ photons. For the orange light of wavelength $\lambda = 610 \text{ nm}$, how much energy is contained in the pulse?

Solution

The energy of a photon (massless particle moving at the speed of light) is given by

$$E_\gamma = hf = \frac{hc}{\lambda},$$

therefore for N_γ photons

$$E_{\text{pulse}} = N_\gamma \frac{hc}{\lambda} = 203 \text{ eV},$$

which can be calculated using the values $h = 4.135\,667\,7 \times 10^{-15} \text{ eV s}$ and $c = 2.9979 \times 10^8 \text{ m/s}$, or analogously using $hc = 1.240 \times 10^3 \text{ eV nm}$.

• **Photoelectric effect / Ex. 3.35**

What is the maximum wavelength λ_{max} of incident light that can produce photoelectrons from silver, which has work function $\phi = 4.64 \text{ eV}$? What will be the maximum kinetic energy of the photoelectrons if the wavelength is halved, i.e., $\lambda' = \lambda_{\text{max}}/2$?

Solution

The work function ϕ is the *minimum* energy required to remove an electron from the surface of a material. Therefore, in order to produce a photoelectron, the photon energy must fulfill the condition

$$E_\gamma \geq \phi$$

The wavelength of the incident light is inversely proportional to the energy

$$\lambda = \frac{c}{f} = \frac{hc}{hf} = \frac{hc}{E_\gamma}.$$

Therefore, the condition on the wavelength reads

$$\lambda \leq \frac{hc}{\phi}$$

which using $hc = 1240 \text{ eV nm}$ yields

$$\lambda_{\text{max}} = \frac{hc}{\phi} = 267.2 \text{ nm}.$$

The energy available for the photoelectron is the amount that exceeds the work function. In the non-relativistic limit, the maximum kinetic energy is given by

$$K_{\text{max}} = E_\gamma - \phi.$$

Therefore, if $\lambda' = \lambda_{\text{max}}/2$, then

$$E' = \frac{hc}{\lambda'} = 2 \frac{hc}{\lambda_{\text{max}}} = 2\phi$$

yielding

$$K_{\text{max}} = E'_\gamma - \phi = \phi = 4.64 \text{ eV}.$$

• **Diffraction/ Ex. 3.10**

Light from a slit passes through a transmission diffraction grating of 400 lines/mm, which is located at $\ell = 3.0$ m from a screen. What are the distances on the screen (from the unscattered slit image) of the three brightest visible (first-order) hydrogen lines?

Solution

The distance between the rulings is given by $d = (400 \text{ mm}^{-1})^{-1} = 2.5 \mu\text{m}$. At the first order ($n = 1$) the diffraction angle is given by

$$\sin(\theta) = \frac{\lambda}{d}.$$

The diffracted light beam has components $x = \ell$ and y related by

$$\tan(\theta) = \frac{y}{x}.$$

The distance along the screen from the unscattered slit image is simply the y component

$$y = x \tan \left[\sin^{-1} \left(\frac{\lambda}{d} \right) \right]$$

yielding

- $\lambda = 656.5 \text{ nm}$: $y = 0.816 \text{ m}$
- $\lambda = 486.3 \text{ nm}$: $y = 0.595 \text{ m}$
- $\lambda = 434.2 \text{ nm}$: $y = 0.529 \text{ m}$

• **Blackbody radiation / Ex. 3.17**

Calculate $\lambda_{\max}$ for blackbody radiation for (a) liquid helium ($T = 4.2\text{ K}$), (b) room temperature ($T = 293\text{ K}$), (c) a steel furnace ($T = 2500\text{ K}$), and (d) a blue star ($T = 9000\text{ K}$).

Solution

For a blackbody at temperature T the maximum of the distribution is found for

$$\lambda_{\max} T = 2.989 \times 10^{-3} \text{ mK},$$

hence $\lambda_{\max}$ is inversely proportional to T , yielding

$$(a) \lambda_{\max} = 0.69 \times 10^{-3} \text{ m}$$

$$(b) \lambda_{\max} = 9.89 \times 10^{-6} \text{ m}$$

$$(c) \lambda_{\max} = 1.16 \times 10^{-6} \text{ m}$$

$$(d) \lambda_{\max} = 0.322 \times 10^{-6} \text{ m}$$

• **Human radiation / Ex. 3.20**

(a) At what wavelength will the human body radiate the maximum radiation? (b) Estimate the total power radiated by a person of medium build (assume an area given by a cylinder of height $h = 175\text{ cm}$ and radius $r = 13\text{ cm}$). (c) Using your answer to (b), compare the energy radiated by a person in one day with the energy intake of a $\mathcal{E} = 2000\text{ kcal}$ diet.

Solution

(a) At an average temperature of $T = 37^\circ\text{C}$ or $T = 310\text{ K}$ the maximum of the distribution is found for

$$\lambda_{\max} = \frac{1}{T} 2.989 \times 10^{-3} \text{ m K} = 9.35 \times 10^{-6} \text{ m}.$$

(b) Assuming a blackbody radiation ($\epsilon = 1$), the power per unit area is given by

$$R = \sigma T^4 = 524 \text{ W/m}^2$$

where $\sigma = 5.6704 \times 10^{-8} \text{ W/(m}^2\text{K}^4)$ is the Stefan–Boltzmann constant, and for an area of the cylinder

$$A = 2\pi r(r + h) = 1.54 \text{ m}^2$$

the total power is

$$P = RA = 807 \text{ W}.$$

(c) For the last comparison, one needs to convert calories to Joules

$$1 \text{ cal} = 4.184 \text{ J},$$

and obtain that in a day there are $86\,400\text{ s}$. Hence, The total energy is

$$\mathcal{E}_{\text{cylinder}} = P \cdot t = 6.97 \times 10^7 \text{ J} > \mathcal{E}_{\text{diet}} = 8.37 \times 10^6 \text{ J}.$$

Note that the cylinder may be a crude approximation of the area of an average human body and that hair covering and clothing minimize radiation. Moreover, this calculation does not take into account the radiation emitted from the environment.

• **Line spectra of H emitted by a moving star / Ex. 3.14**

Suppose that a detector in a new space telescope will be capable of detecting visible light in the wavelength range of $\lambda_{\min} = 400 \text{ nm}$ to $\lambda_{\max} = 700 \text{ nm}$. (a) List all the wavelengths for the hydrogen atom that are in this range and their series name. (b) The detector measures visible wavelengths of $\lambda_1 = 537.5 \text{ nm}$, $\lambda_2 = 480.1 \text{ nm}$, and $\lambda_3 = 453.4 \text{ nm}$, that researchers believe are due to the hydrogen atom. Why are all the known visible hydrogen lines not detected? (c) Use these data to calculate the speed of the stellar object that emitted the spectra. Assume that the object is not rotating. Why might rotation be an issue?

Solution

(a) The wavelengths of the hydrogen atom in the visible range are the first four of the Balmer series ($n = 2$) given by

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n^2} - \frac{1}{k^2} \right)$$

for $n = 2$ and $n < k \leq 6$ one indeed finds

- $k = 3$: $\lambda = 656.5 \text{ nm}$ (red)
- $k = 4$: $\lambda = 486.3 \text{ nm}$ (blue-green)
- $k = 5$: $\lambda = 434.2 \text{ nm}$ (violet)
- $k = 6$: $\lambda = 410.3 \text{ nm}$ (violet)

while for $k = 7$ the next line is at $\lambda = 397.1 \text{ nm} < 400 \text{ nm}$. The wavelengths of the Lyman series ($n = 1$) are in the ultraviolet region and those of the Paschen series ($n = 3$) are in the infrared region.

(b) In principle, the $k = 6$ line of hydrogen should also fall within the visible range. This means that the source emitting the light is moving. One can find the velocity of the source by calculating the *redshift* of the experimentally observed lines with respect to their theoretical values. Indeed, the observed lines are shorter than the theoretical ones, meaning that the corresponding frequency is higher and that the source is therefore moving away from the observer.

(c) Using the formula of the relativistic Doppler effect for the wavelength

$$\frac{\lambda_{\text{observed}}}{\lambda_{\text{source}}} = \sqrt{\frac{1 + \beta}{1 - \beta}}.$$

Solving for β one gets

$$\beta = \frac{\left(\frac{\lambda_{\text{observed}}}{\lambda_{\text{source}}} \right)^2 - 1}{\left(\frac{\lambda_{\text{observed}}}{\lambda_{\text{source}}} \right)^2 + 1}.$$

calculating this for, e.g., $k = 3$, one obtains $\beta = 0.0995 \approx 0.10$ or $v \approx 0.10c = 3.0 \times 10^7 \text{ m/s}$. A similar result is obtained with any other pair of wavelengths.

Note that, due to the Doppler shift, the $k = 7$ and $k = 8$ would also fall within the detection range of the telescope.

If the object were rotating, one side would move towards the observer and the other side away from the observer. Moreover, while the angular ω velocity is constant, the linear velocity of a point on the surface depends on the distance from the rotation axis, and therefore varies with latitude ($v = \omega r$). This would result in a range of wavelengths observed by the detector.

4. STRUCTURE OF THE ATOM

• **Rutherford scattering / Ex. 4.11**

The nuclear radii of aluminum and gold are approximately $r_{\text{Al}} = 3.6 \text{ fm}$ and $r_{\text{Au}} = 7.0 \text{ fm}$, respectively. The radii of protons and alpha particles are $r_{\text{p}} = 1.3 \text{ fm}$ and $r_{\alpha} = 2.6 \text{ fm}$, respectively.

(a) What energy α particles would be needed in head-on collisions for the nuclear surfaces to just touch? —This is about where the nuclear force becomes effective, it can be neglected at larger distances, and consider only the Coulomb interaction.

(b) What energy protons would be needed?

For both (a) and (b), perform the calculation for aluminum and for gold.

Solution

In all cases, the kinetic energy of the incoming particle is transformed into potential energy

$$K = -\Delta V = |V| = \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 (r_1 + r_2)},$$

where r_1 and r_2 are the radius of the incoming particle and the nuclear radius, respectively, so that $r_1 + r_2$ is the minimum distance in the process.

For the proton $Z_{\text{p}} = 1$, an α particle (He^{2+}) is composed by two protons and two neutrons, so that $Z_{\alpha} = 2$. For aluminum $Z_{\text{Al}} = 13$ and for gold $Z_{\text{Au}} = 79$.

Therefore:

$$\begin{aligned} \text{(a) Al: } K &= 6.04 \text{ MeV} \\ \text{Au: } K &= 23.7 \text{ MeV} \end{aligned}$$

$$\begin{aligned} \text{(b) Al: } K &= 3.82 \text{ MeV} \\ \text{Au: } K &= 13.7 \text{ MeV} \end{aligned}$$

5. WAVE PROPERTIES OF MATTER AND QUANTUM MECHANICS I

• Bragg scattering / Ex. 5.19

In an X-ray scattering experiment, an intense reflected beam is found at $\phi = 53^\circ$ for a crystal with an interatomic distance of $D = 0.23 \text{ nm}$, see the sketch (assuming for the sake of simplicity that $D_x = D_y = D$). What is the lattice spacing of the planes responsible for the scattering? Assuming first-order diffraction, what is the wavelength λ , frequency f , and energy E of the incident photons?

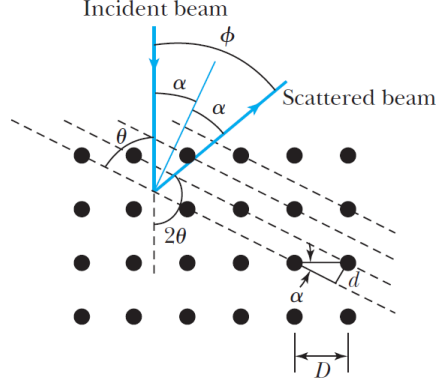


Figure 2: Bragg diffraction

Solution

A schematic of the electron scattering is reproduced in the Figure.

The distance between scattering planes (dashed lines) in terms of the interatomic distance is given by

$$d = D \sin(\alpha) = D \sin(\phi/2) = 0.103 \text{ nm}$$

The wavelength of the incident light is given by

$$n\lambda = 2d \sin(\theta_n)$$

where θ_n is the diffraction angle of the n -th order diffracted wave, so that for $n = 1$ $\theta_1 = \pi/2 - \phi/2 = \pi/2 - \alpha$. From this:

$$\lambda = 2d \sin \theta = 2D \sin \alpha \cos \alpha = D \sin 2\alpha = D \sin \phi = 0.184 \text{ nm}.$$

Given the wavelength, the frequency is easily obtained as

$$f = \frac{c}{\lambda} = 1.63 \times 10^{18} \text{ Hz},$$

from which the energy

$$E = hf = \frac{hc}{\lambda} = 6739 \text{ eV}$$

where $hc = 1240 \text{ eV nm} = 1240 \times 10^{-9} \text{ eV m}$.